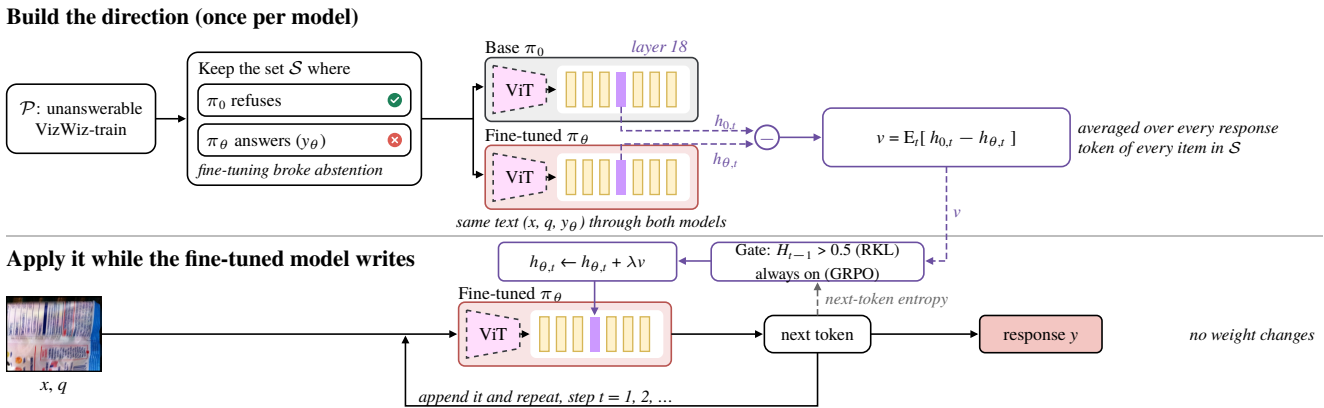

Build the direction (once per model)
$\mathcal{P}$: unanswerable VizWiz-train
Keep the set $\mathcal{S}$ where
π_0 refuses
π_θ answers (y_θ)
fine-tuning broke abstention
Base π_0
layer 18
ViT
Fine-tuned π_θ
ViT
same text (x, q, y_θ) through both models
$h_{0,t}$
$h_{\theta,t}$
$v = \mathrm{E}_t[\, h_{0,t} - h_{\theta,t} \,]$
averaged over every response token of every item in $\mathcal{S}$
v
Apply it while the fine-tuned model writes
$h_{\theta,t} \leftarrow h_{\theta,t} + \lambda v$
Gate: $H_{t-1} > 0.5$ (RKL)
always on (GRPO)
next-token entropy
Fine-tuned π_θ
ViT
next token
response y
no weight changes
x, q
append it and repeat, step t = 1, 2, …